



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Summer Examinations 2023/2024

Course Instance Codes	3BA1, 4BA161, 4BME1, 4BMU1, 1OA2, 1OA3, 4BCT1, 4BMS2, 4BS2
Exams	3rd and 4th year (Arts and Science), Study Abroad / International Exchange
Module Code	CS402
Module	Cryptography
External Examiner	Prof. C. Roney-Dougal
Internal Examiners	Dr N. Madden *Dr K. Jennings
<u>Instructions:</u>	Answer all four questions. Each question carries 25 marks. All workings must be shown.
Duration	2 hours
No. of Pages	4 pages, including this one
Discipline	Mathematics
<u>Requirements:</u>	No special requirements
Release to Library	Yes
Release in Exam Venue	Yes
MCQ	No
Statistical / Log Tables	No
Non-programmable Calculator	Yes
Mathematical Tables	No
Graph paper	No

The following table may be helpful throughout Q1:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Q	R	S	T	U	V	W	X	Y	Z	␣					
16	17	18	19	20	21	22	23	24	25	26					

Q1. (a) The ciphertext string

JCRYGRY_C

was produced using a shift cipher over the 27-letter alphabet

$$A = 0, B = 1, \dots, Z = 25, _ = 26,$$

where the final character $_$ indicates a space.

By conducting a frequency analysis of a larger sample of ciphertexts, it has been found that the plaintext symbol $_$ corresponds to the ciphertext symbol Y. Determine the plaintext string corresponding to the ciphertext string given above. [5 marks]

(b) Identify $\mathbf{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{A, \dots, Z, _ \}$ as in part (a). The ciphertext string ZLV $_$ was obtained using a 2-dimensional Hill cipher over $\mathbf{Z}_{27}$ with key

$$\begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}.$$

Find the corresponding plaintext. [10 marks]

(c) Briefly describe how the Vigenère cipher works. Using the key ‘FAB’ and the same 27 character alphabet as described in parts (a) and (b) above encrypt the plaintext ‘BEATLES’ with a Vigenère cipher. [10 marks]

Q2. (a) Formally define what is meant by a (symmetric) *cryptosystem*. [5 marks]

(b) Briefly describe what a *perfectly secure* (symmetric) cryptosystem is. Do perfectly secure cryptosystems exist? Justify your answer. [5 marks]

(c) State and prove Shannon’s Theorem on perfect security (perfect secrecy). [15 marks]

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- Q3.** (a) Describe the difference between *symmetric cryptosystems* and *public-key cryptosystems*. Give an example of each type of cryptosystem along with an advantage and a disadvantage of each type. [5 marks]
- (b) Describe the key generation, encryption, and decryption steps of the *RSA* cryptosystem. [5 marks]
- (c) Describe the method of *fast exponentiation* for computing $a^e \bmod n$, where $a, e, n \in \mathbf{Z}$ with $e, n \geq 1$. [5 marks]
- (d) Bob claims that if an efficient algorithm for deciding whether an integer is prime were ever to be developed, then the RSA cryptosystem would no longer be secure. Do you agree with Bob? Justify your answer. [5 marks]
- (e) Bob claims that since 1105 is a Carmichael number, the Fermat primality test cannot detect that 1105 is composite. Alice disagrees and argues that for any composite n , a random choice of $a \in \{1, \dots, n-1\}$ has a (possibly small) chance of satisfying $\gcd(a, n) > 1$, in which case $a^{n-1} \not\equiv 1 \pmod{n}$. With whom (if anyone) of the two do you agree? Justify your answer. [5 marks]

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- Q4.** (a) A linear feedback shift register $\mathcal{L}$ of length 2 is used to encrypt messages using a stream cipher. It is known that when the cipher is used to encrypt the plaintext string $(x_0, x_1, x_2, x_3) = (0, 1, 1, 1)$, then the corresponding ciphertext string is $(y_0, y_1, y_2, y_3) = (1, 1, 1, 0)$. Find the connection polynomial of $\mathcal{L}$. [5 marks]
- (b) A linear feedback shift register of length 4 with connection polynomial

$$C(X) = 1 + X + X^2 + X^4$$

over $\mathbf{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = s_2 = 1$ and $s_1 = s_3 = 0$. Determine the least period of the resulting infinite sequence $s_0, s_1, \dots$. [5 marks]

- (c) Define what it means for a group to be *cyclic*. State (without proof) Gauss's characterisation of those $n \geq 1$ for which the group $\mathbf{Z}_n^\times$ is cyclic. Is $\mathbf{Z}_{315}^\times$ cyclic? Justify your answer. [5 marks]
- (d) Alice and Bob use the cyclic group $\mathbf{Z}_{29}^\times$ and its generator $g = 2$ to perform a Diffie-Hellman key exchange. Alice secretly chooses $a = 5$ and Bob secretly chooses $b = 4$. Determine their shared key (as an element of $\mathbf{Z}_{29}^\times$). [5 marks]
- (e) Let E be the set of points of the elliptic curve over $\mathbf{R}$ defined by the equation

$$y^2 = x^3 + 3x,$$

with point at infinity $\mathcal{O}$.

- i. Find all points $P \in E$ that satisfy $P + P = \mathcal{O}$.
- ii. Find a point $Q \in E$ with $Q + Q \neq \mathcal{O}$. [5 marks]